# MIDTERM EXAMINATION (Spring 2011)
# STA301- Statistics and Probability

**Question No: 1 (Marks: 1) - Please choose one**
**A five number summery consist of :**

► $X_0, Q_1, Median, Q_3, and X_m$ (Page 97)

► $X_m, Q_1, Mean, Q_3, and\ X_0$

► $X_m, Q_1, Mode, Q_3, and\ X_0$

► $X_0, Q_1, Median, Q_2, and X_m$

**Question No: 3 (Marks: 1) - Please choose one**
in a multiplication theorem $P(A \cap B)$ equals:

►P(A) P(B)
►P(A) + P(B)
**►P(A) * P(B|A) (Page 159)**
►P(B\A) * P(B)

**Question No: 4 (Marks: 1) - Please choose one**
The probability of drawing 'White' ball from a bag containing 4 red , 8 black and 3 white ball is
►0
**►3/15**
►1/12
►1/2

**Question No: 6 (Marks: 1) - Please choose one**
If *A* and *B* are mutually exclusive events then *P*(*A* or *B*) equals:
► *P*(*A*) + *P*(*B*) - *P*(*A* and *B*)
► *P*(*A*) * *P*(*B*)
**► *P*(*A*) + *P*(*B*) (Page 155)**
► *P*(*A*|*B*) + *P*(*B*|*A*)

دنیا میں سب سے مشکل کام اپنی اصلاح اور سب سے آسان کام دوسروں پر نکتہ چینی کرنا ہے





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**Question No: 7 (Marks: 1) - Please choose one**
The simultaneous occurrence of two events is called
► Prior probability
► Subjective probability not confirmed
► Conditional probability
► **Joint probability** click here for detail

**Question No: 8 (Marks: 1) - Please choose one**
First moment about origin is always equals to:
► **Mean (Page 119)**
► Variance
► Standard Deviation
► Zero

**Question No: 9 (Marks: 1) - Please choose one**
Which one of the following measurement does ***not*** divide a set of observations into equal parts?

► Quartiles
► Deciles
► Percentiles
► **Standard deviations**

**Question No: 10 (Marks: 1) - Please choose one**
**if first and third quartile are 21.16 and 56.36 respectively, then the quartile deviation is:**
► **17.1 (Page 84)**
► 34.2
► 51.3
► 50.5

**Question No: 11 (Marks: 1) - Please choose one**
**The height of student is 60 inch. This is an example of……?**
► Continues data
► Qualitative data
► Categorical data
► **Discrete data**

**Question No: 12 (Marks: 1) - Please choose one**
Which of the statement is true regarding a sample?
► **It is a part of population (Page 13)**
► It must contain at least five observations
► It refers to descriptive statistics
► It produces true value

**Question No: 13 (Marks: 1) - Please choose one**
Which one of the following graphs is used for a time series data?

► Histogram
► Historigram
► Frequency curve
► **Frequency polygon** Click here for detail

**Question No: 14 (Marks: 1) - Please choose one**
**Which of the following comes first to make frequency distribution**

► No. of groups
► Class intervals
► **Rang (Page 28)**
► Tally marks

**Question No: 15 (Marks: 1) - Please choose one**
**The average which is defines as reciprocal of arithmetic mean of the reciprocals of the values is called**

► Geometric Mean
► **Harmonic mean (Page 77)**
► Mode
► Median

**Question No: 16 (Marks: 1) - Please choose one**
If a car is running at the rate of 15km/hr during first 30km, at 20km/hr during the second 30, which type of average will be used to find the average speed/hr:

► **Weighted mean (Not sure)**
► Harmonic mean
► Arithmetic mean
► Geometric mean



# MIDTERM EXAMINATION (Spring 2011)
# STA301- Statistics and Probability

**Question No: 1 (Marks: 1) - Please choose one**
The probability of drawing a 'jack card 'from 52 playing cards is:

► $\frac{1}{52}$
► $\frac{4}{52}$
► $\frac{13}{52}$
► $\frac{26}{52}$

**Question No: 2 (Marks: 1) - Please choose one**
If all the values fall on the same straight line and the line has a positive slope then what will be the value of the correlation coefficient '**r**':

► $0 \leq r \leq 1$
► $r \geq 0$
► **r = +1**
► r=-1

**Question No: 3 (Marks: 1) - Please choose one**
If a curve has a longer tail to the right, it is called:

► **Positively skewed (Page 39)**
► Negatively skewed
► J-shaped
► Symmetric

**Question No: 04 (Marks: 1) - Please choose one**
**Which one of the following is not included in measures of central tendency?**

► **Quartile deviation (Page 82)**
► Harmonic mean
► Geometric mean
► Arithmetic mean







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**Question No: 05 ( Marks: 1 ) - Please choose one**
Which of the following is not based on all the observations?

► Arithmetic Mean
► Geometric Mean
► Harmonic mean
► **Mode**

**Question No: 06 ( Marks: 1 ) - Please choose one**
What is the Standard Deviation of 7, 7, 7, 7, 7, 7, 7

►49
►1
► **0** **Standard deviation will always be zero if all the values in data are same**
►7

## MIDTERM EXAMINATION (Spring 2010)
## STA301- Statistics and Probability (Session - 4)

**Question No: 1 (Marks: 1) - Please choose one**
10! =…….

► 362880
► **3628800**
► 362280
► 362800

**Question No: 2 (Marks: 1) - Please choose one**
If a player well shuffles the pack of 52 playing cards, then the probability of a black card from 52 playing cards is:

►1/52
► **13/52**
►4/52
► **26/52**





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**Question No: 3 (Marks: 1) - Please choose one**
The probability of drawing a 'jack card 'from 52 playing cards is:

- ► 1/52
- ► **4/52**
- ► 13/52
- ► 26/52

**Question No: 4 (Marks: 1) - Please choose one**
Which dispersion is used to compare variation of two series?

- ► **C.V. (Page 93)**
- ► Q.D.
- ► M.D.
- ► S.D.

**Question No: 5 (Marks: 1) - Please choose one**
If all the values fall on the same straight line and the line has a positive slope then what will be the value of the correlation coefficient '**r**':

- ► $0 \leq r \leq 1$
- ► $r \geq 0$
- ► **r = +1 (Page 129)**
- ► r=-1

**Question No: 6 (Marks: 1) - Please choose one**
In a regression line Y= a + bX, the value of the correlation coefficient will be zero if:

- ► **Intercept a = 0**
- ► Intercept $a \neq 0$
- ► Slope b = 0
- ► Slope $b \neq 0$

**Question No: 7 (Marks: 1) - Please choose one**
When two coins are tossed the probability of at least one head is:

- ► 1/4
- ► **3/4 (HH, HT, TH, TT)**
- ► 2/4
- ► 4/4





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**Question No: 8 (Marks: 1) - Please choose one**
Which one of the following measurement does ***not*** divide a set of observations into equal parts?

- ► Quartiles
- ► Deciles
- ► Percentiles
- ► **Standard deviations rep**

**Question No: 9 (Marks: 1) - Please choose one**
In the model Y = mX + a, Y is also known as the:

- ► Predictor variable
- ► Independent variable
- ► **Predicted (dependent) variable (Page 121)**
- ► Explanatory variable

**Question No: 10 (Marks: 1) - Please choose one**
According to empirical rule approximately 95% of the measurements will fall under which interval?

- ► $\bar{X} \pm S$
- ► $\bar{X} \pm 2S$ **correct**
- ► $\bar{X} \pm 3S$
- ► $\bar{X} \pm 4S$

**Question No: 11 (Marks: 1) - Please choose one**
Which one of the following is written at the top of the table?

- ► Source note
- ► Foot note
- ► Prefatory note
- ► **Title** **Click here for detail**

**Question No: 12 (Marks: 1) - Please choose one**
If a curve has a longer tail to the right, it is called:

- ► **Positively skewed rep**
- ► Negatively skewed
- ► J-shaped
- ► Symmetric





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**Question No: 13 (Marks: 1) - Please choose one**
Which one of the following is the class frequency?

- ► **The number of observations in each class**
- ► The difference between consecutive lower class limits
- ► Always contains at least 5 observations
- ► Usually a multiple of the lower limit of the first class

**Question No: 14 (Marks: 1) - Please choose one**
If X is a discrete random variable, then the function $f(x)$ is

- ► A probability function
- ► A probability density function
- ► A density function
- ► **A distribution function (Page 173)**

**Question No: 15 (Marks: 1) - Please choose one**
Which one of the following graphs is used for a time series data?

- ► **Histogram**
- ► Historigram
- ► Frequency curve
- ► **Frequency polygon**

**Question No: 16 (Marks: 1) - Please choose one**
If you connect the mid-points of rectangles in a histogram by a series of lines that also touches the x-axis from both ends, what will you get?

- ► Ogive
- ► Frequency polygon
- ► **Frequency curve (Page 38)**
- ► Historigram

**Question No: 17 (Marks: 1) - Please choose one**
If mean of the two observations is 10.5, then median of these two observations will be:

- ► 7.5
- ► 8.5
- ► 9.5
- ► **10.5 (Page 68)**



**Question No: 18 (Marks: 1) - Please choose one**
Which one is the formula of mid range?

- ► $x_m - x_0$
- ► $x_0 - x_m$
- ► $\frac{x_0 - x_m}{2}$
- ► $\frac{x_0 + x_m}{2}$ **(Page 80)**

**Question No: 19 (Marks: 1) - Please choose one**
Which one of the following is not included in measures of central tendency?

- ► **Quartile deviation (page 82)**
- ► Harmonic mean
- ► Geometric mean
- ► Arithmetic mean

**Question No: 20 (Marks: 1) - Please choose one**
For the given data 2, 3, 7, 0, -8 G. M will be:

- ► Negative
- ► Positive
- ► **Zero (Page 75)**
- ► Undefined

## MIDTERM EXAMINATION (Spring 2010)
## STA301- Statistics and Probability

**Question No: 1 (Marks: 1) - Please choose one**
If a player well shuffles the pack of 52 playing cards, then the probability of a black card from 52 playing cards is:

- ► 1/52
- ► 13/52
- ► 4/52
- ► **26/52** **rep**

عقل مند کہتا ہے میں کچھ نہیں جانتا جبکہ بے وقوف کہتا ہے کہ میں سب کچھ جانتا ہوں

**Question No: 2 (Marks: 1) - Please choose one**
The probability of drawing a 'jack card 'from 52 playing cards is:

► 1/52
► 13/52
► **4/52 rep**
► 26/52

**Question No: 3 (Marks: 1) - Please choose one**
In a regression line Y= a + bX, the value of the correlation coefficient will be zero if:

► **Intercept a = 0 rep**
► Intercept a ≠ 0
► Slope b = 0
► Slope b ≠ 0

**Question No: 4 (Marks: 1) - Please choose one**
Which one of the following measurement does ***not*** divide a set of observations into equal parts?

► Quartiles
► Deciles
► Percentiles
► **Standard deviations rep**

**Question No: 5 (Marks: 1) - Please choose one**
Which one of the following graphs is used for a time series data?

► Histogram
► Historigram
► Frequency curve
► **Frequency polygon rep**

**Question No: 6 (Marks: 1) - Please choose one**
If you connect the mid-points of rectangles in a histogram by a series of lines that also touches the x-axis from both ends, what will you get?

► Ogive
► Frequency polygon
► **Frequency curve (Page 38) rep**
► Historigram

خود کو تمھیں سے بڑھ کر کوئی اچھا مشورہ نہیں دے سکتا

**Question No:7 (Marks: 1) - Please choose one**
Which one is equal to explained variation divided by total variation?

►Sum of square due to regression
► **Coefficient of determinant** Click here for detail
►Standard error estimate
► Coefficient of correlation

**Question No: 8 (Marks: 1) - Please choose one**
in the given series 1,2,1,1,2,2,2,3,4,5,3,2,3,1,4,2,3 mode of given is
► 4
► 3
►1
► **2**

**Question No: 9 (Marks: 1) - Please choose one**
True for the population,
► it must be large number of values
► It must refer to people
► **It is collection of individual objects or measurement (Page 12)**
► It is the small part of whole

**Question No:10 (Marks: 1) - Please choose one**
Data arrangement in ascending or descending order
► **Array data (Page 47)**
► Group data
► Ungroup data
► Raw data

**Question No: 11 (Marks: 1) - Please choose one**
What is the main objective of Descriptive statistics?
► To test population properties
► To describe the data we collected
► To infer something about the population
► **Making estimate** **Click here for detail**

**Question No: 12 (Marks: 1) - Please choose one**
Which measure of central tendency?
► Variation of distribution
► **Average of distribution (Page 51)**
► Scattering of distribution
► Dispersion of distribution

جو شخص ناکامیوں سے ڈر کر بھاگتا ہے کامیابی اُس سے ڈر کر بھاگتی ہے

**Question No: 13 (Marks: 1) - Please choose one**
If a=4 b=2 estimate line (i.e y=a+bx) and independent variable has value 3 the value of dependent variable
► 6
► 9
► **10** **4+2(3)=10**
► 11

**Question No: 14 (Marks: 1) - Please choose one**
The number of ways in which 4 books can be arranged
► 4
► 6
► 12
► **24 (Page 142)**

**Question No: 15 (Marks: 1) - Please choose one**
If we plot paired observed (x,y)=1…….n on graph is called,
► Polygon
► Freasito diagram
► Scatter diagram
► Cumulative frequency diagram

**Question No: 16 (Marks: 1) - Please choose one**
The simultaneous occurrence of two events is called
► Descriptive probability
► Subjective probability
► Conditional probability
► **Joint probability**

**Question No: 17 (Marks: 1) - Please choose one**
Which one is the not measure of dispersion.
► The range
► **50th percentile**
► Inter quartile range
► Variance

**Question No: 18 (Marks: 1) - Please choose one**
In positively skew cure which relation is
► The mean, median and mode are equal
► **Mean is greater then median** **Click here for detail**
► Median is greater then mean
► Standard deviation must be greater then mean or median

جو لوگوں کے سامنے فخر کرتا ہے وہ لوگوں کی نظروں سے گر جاتا ہے

**Question No: 19 (Marks: 1) - Please choose one**
When coin tossed we get only
► **1 outcome**
► 2 outcomes
► 3 outcomes
► 4 outcomes

**Question No: 20 (Marks: 1) - Please choose one**
When mean is 25 and S.D is 5 then CV is

► 100%
► 25%
► **20%**
► 10%

# MIDTERM EXAMINATION
## Spring 2009
## STA301- Statistics and Probability (Session - 1)

**Question No: 1 ( Marks: 1 ) - Please choose one**
For a positively skewed distribution $m_3$ will be:
► **Positive** **(Page 119)**
► Negative
► Zero
► 1

**Question No: 2 ( Marks: 1 ) - Please choose one**
When data is labeled to identify an attribute of element, the measurement scale is:
► **Ordinal** **(Page 9)**
► Interval
► Nominal
► Ratio





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**Question No: 3 ( Marks: 1 ) - Please choose one**

Suppose the estimated equation is $\hat{Y} = 5 - 2X$ has been calculated for a set of data. What is slop of the line:

- ► 0
- ► 2
- ► **-2 (Page 121)**
- ► 5

**Question No: 4 ( Marks: 1 ) - Please choose one**

If P(B|A) = 0.25 and $P(A \cap B) = 0.20$, then P(A) is:

- ► 0.05
- ► **0.80 (Page 159)**
- ► 0.95
- ► 0.75

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)}$$

$$= P(A) = \frac{P(A \cap B)}{P(B \mid A)}$$

$$= \frac{0.20}{0.25} = 0.80$$

**Question No: 5 ( Marks: 1 ) - Please choose one**

Which branch of statistics deals with the techniques that are used to organize, summarize, and present the data:

- ► Advance statistics
- ► Probability statistics
- ► **Descriptive statistics (Page 61)**
- ► Inferential statistics

**Question No: 6 ( Marks: 1 ) - Please choose one**

In a sample of 800 students in a university, 160, or 20%, are Business majors. Based on the above information, the school's paper reported that "20% of all the students at the university are Business majors." This report is an example of :

- ► A sample
- ► A population
- ► Statistical inference
- ► **Descriptive statistics**







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**Question No: 7 ( Marks: 1 ) - Please choose one**
A set that contains all possible outcomes of a system is known as

- Finite Set
- Infinite Set
- **Universal Set (Page 134)**
- No of these

**Question No: 8 ( Marks: 1 ) - Please choose one**
If X and Y are independent, then Var(X-Y) is equal to:

- $Var(X)-Var(Y)$
- $Var(X)+Var(Y)$
- $Var(X+Y)$
- Zero

**Question No: 9 ( Marks: 1 ) - Please choose one**
Which of the following is the class frequency

- **The number of observations in each class**
- The difference between consecutive lower class limits
- Always contains at least 5 observations
- Usually a multiple of the lower limit of the first class

**Question No: 10 ( Marks: 1 ) - Please choose one**
How to construct the class interval:

- Divide the class frequencies in half
- Divide the class frequency by the number of observations
- **Find the difference between consecutive lower class limits**
- Count the number of observations in the class

**Question No: 11 ( Marks: 1 ) - Please choose one**
Data in the Population Census Report is:

- **Ungrouped data (Page 11)**
- Secondary data
- Primary data
- Arrayed data





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**Question No: 12 ( Marks: 1 ) - Please choose one**
What is the range of -2,-3,-5,-10 :

► **-12 (Page 28)**
► 8
►-8
► 2

Question No: 13 ( Marks: 1 ) - Please choose one
The algebraic sum of deviations from mean is:
► Maximum
► Minimum
► **Zero (Page 86)**
► Undefined

Question No: 14 ( Marks: 1 ) - Please choose one
The sum of squares of deviations from mean is:

► Undefined
► Zero
► **Maximum**
► Minimum

Question No: 15 ( Marks: 1 ) - Please choose one
Statistic is a numerical quantity, which is calculated from:

► Population
► **Sample (Page 7)**
► Data
► Observations

Question No: 16 ( Marks: 1 ) - Please choose one
Which of the following is not based on all the observations?
► Arithmetic Mean
► Geometric Mean
► Harmonic mean
► **Mode (rep)**



# MIDTERM EXAMINATION
## Spring 2009
## STA301- Statistics and Probability (Session - 1)

**Question No: 1 ( Marks: 1 ) - Please choose one**
If any value in the data is zero, then it is not possible to have:
► Harmonic Mean
► Arithmetic Mean
► Median
**► Mode (rep)**

**Question No: 2 ( Marks: 1 ) - Please choose one**
For a symmetrical distribution… .....is equidistance from median:
► X0 and Xm
**► Q1 and Q3 (Page 97)**
► X0 and Q1
► Xm and Q3

**Question No: 3 ( Marks: 1 ) - Please choose one**
Which one of the following measure is not used in 'five number summery':
□ $X_0$
□ $Q_3$
□ $\overline{X}$ (*Page* 97)
□ $Q_1$

**Question No: 4 ( Marks: 1 ) - Please choose one**
If Mean = 25 & S.D is 5 then C.V is
► 100%
► 25%
**► 20% (Page 93)**
► 10%

**Question No: 5 ( Marks: 1 ) - Please choose one**
A coin is tossed 3 times then, the number of sample points ii the sample space is:
►3
►8
**►6 (Page 145)**
►4

بہترین تجربہ وہ ہے جس سے نصیحت حاصل ہو




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**Question No: 6 ( Marks: 1 ) - Please choose one**
What is the difference between a permutation and a combination:
► **In a permutation order is important and in a combination it is not** Click here for detail
► In a permutation order is not important and in a combination it is important
► A combination is based on the classical definition of probability
► A permutation is based on the classical definition of probability

**Question No: 7 ( Marks: 1 ) - Please choose one**
What we consider in simple correlation analysis:
► Several independent variables
► **Strength of the association between two variables (Page 128)**
► Intercept with the X-axis
► Intercept with the *Y*-axis

**Question No: 8 ( Marks: 1 ) - Please choose one**
If all the values fall on the same straight line and the line has a positive slope then what will be the value of the correlation coefficient 'r':
► 0 = r = 1
► r = 0
► **r = +1 (rep)**
► r=-1

**Question No: 9 ( Marks: 1 ) - Please choose one**
Which of the following measure of dispersion is least affected by extreme values of observations in a data:
► **Range Page 82)**
► Quartile deviation
► Mean absolute deviation
► Standard deviation

**Question No: 10 ( Marks: 1 ) - Please choose one**
The approximate number of observations in a set of data covered by the interval, $Median \pm Q.D$ are:
► **50 per cent (Page 84)**
► 68.5 per cent
► 95.4 per cent
► 99 per cent

**Question No: 11 ( Marks: 1 ) - Please choose one**
The dispersion which is calculated from all observed values is:
► **Standard deviation (Page 87)**
► Quartile deviation
► Rang
► Coefficient of Rang

خوبصورتی علم وادب سے ہوتی ہے لباس وحسن سے نہیں

**Question No: 12 ( Marks: 1 ) - Please choose one**
In a set of 20 values all the values are 2, what is the value of Geometric Mean:
►**2** **(Page 75)**
►5
►10
►20

**Question No: 13 ( Marks: 1 ) - Please choose one**
In a set of 10 values all the values are 5, what is the value of 5th Decile:

►2
►**5**
►10
►20

**Question No: 14 ( Marks: 1 ) - Please choose one**
If a car is running at the rate of 15km/hr during first 30km, at 20km/hr during the second 30, which type of average will be used to find the average speed/hr:

►**W.M**
►H.M
►A.M
►G.M

**Question No: 15 ( Marks: 1 ) - Please choose one**
In stem and leaf plot, data is measured on:

►Ratio Scale
►Interval Scale
►Ordinal scale
►**Nominal Scale**

**Question No: 16 ( Marks: 1 ) - Please choose one**
In statistics, we deal with:

►Individuals
►Isolated items
►**Aggregates of facts** **(Page 7)**
►Qualitative data



# MIDTERM EXAMINATION
## Spring 2009
## STA301- Statistics and Probability (Session - 1)

**Question No: 1 ( Marks: 1 ) - Please choose one**
Which of the following is a systematic arrangement of data into rows and columns:
► Component bar chart
► Classification
**► Tabulation**
► Bar chart

**Question No: 2 ( Marks: 1 ) - Please choose one**
For any number k.....1, at least 1 – 1/k2 of the data-values fall within k standard deviations of the mean:
**► Greater than 1 (Page 97)**
► Less than 1
► Greater or equal to 1
► Less or equal to 1

**Question No: 3 ( Marks: 1 ) - Please choose one**
If Mean = 25 & S.D is 5 then C.V is
► 100%
► 25%
**► 20% (Page 93)rep**
► 10%

**Question No: 4 ( Marks: 1 ) - Please choose one** When E is an impossible event,then P(E) is:
**► 0 (Page 146)**
► 0.5
► 1
► 2

**Question No: 5 ( Marks: 1 ) - Please choose one**
The data for an ogive is found in which distribution:
**► A cumulative frequency distribution (Page 43)**
► A relative frequency distribution
► A frequency distribution
► A joint frequency distribution

انسان کے لئے بری صحبت سے بڑھ کر بری کوئی چیز نہیں

**Question No: 6 ( Marks: 1 ) - Please choose one**
Which of the following statements is true regarding a sample:

► **It is a part of population (Page 13) rep**
► It must contain at least five observations
► It refers to descriptive statistics
► It produces True value

**Question No: 7 ( Marks: 1 ) - Please choose one**
Which level of measurement is required for the median:

► Nominal
► **Ordinal**
► Interval
► Ratio

**Question No: 8 ( Marks: 1 ) - Please choose one**
In a the regression line **Y= a + bX+ €** the variable which is non-random is:

► **X Page 122)**
►Y
► Both X and Y
► Neither X nor Y

**Question No: 9 ( Marks: 1 ) - Please choose one**
The variable plotted on the horizontal or X-axis in a scatter diagram is called the:

► **Scatter variable** Click here for detail
► Independent variable
► Dependent variable
► Correlation variable

**Question No: 10 ( Marks: 1 ) - Please choose one**
Which is the formula of range:
► $x_m - x_0$ (*Page* 28)
► $x_0 - x_m$
► $\frac{x_m - x_0}{2}$
► $\frac{x_m + x_0}{2}$







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**Question No: 11 ( Marks: 1 ) - Please choose one**
The descriptive measure of dispersion that is based on the concept of a deviation about the mean is:
► Range
► Inter quartile range
► Quartile deviation
**► Standard deviation**

**Question No: 12 ( Marks: 1 ) - Please choose one**
Which branch of statistics deals with the techniques that are used to organize, summarize, and present the data:
► Advance statistics
► Probability statistics
**► Descriptive statistics (rep)**
► Inferential statistics

**Question No: 13 ( Marks: 1 ) - Please choose one**
A population that can be defined as the aggregate of all the conceivable ways in which a specified event can happen is known as:
► Infinite population
► Finite population
► Concrete population
**► Hypothetical population (Page 12)**

**Question No: 14 ( Marks: 1 ) - Please choose one**
First moment about origin is always equals to:
**► Mean** Click here for detail
► Variance
► Standard Deviation
► Zero

**Question No: 15 ( Marks: 1 ) - Please choose one**
When two dice are rolled, the numbers of possible sample points are:
► 6
► 12
► 24
**► 36 (6*6=36)**

**Question No: 16 ( Marks: 1 ) - Please choose one**
The correlation of coefficient lies between :
► 0 to 1
► 0 *to* ∞
**► - 1 to +1 (Page 128)**
► +1 *to* ∞

جھوٹ رزق کو کھا جاتا ہے

# MIDTERM EXAMINATION
Spring 2009
STA301- Statistics and Probability (Session - 2)

**Question No: 1 ( Marks: 1 ) - Please choose one**
Median can be found only when:
►Data is Discrete
►Data is Attributed  Data is continuous
►Data is continuous  Data is attributed
**►Data is arranged  (Page 52)**

**Question No: 2 ( Marks: 1 ) - Please choose one**
From the following observations 2,3,4,5,4,6,4, the mode is:
►2
►3
**►4**
►5

**Question No: 3 ( Marks: 1 ) - Please choose one**
How to construct the class interval:
►Divide the class frequencies in half
►Divide the class frequency by the number of observations
**►Find the difference between consecutive lower class limits**
►Count the number of observations in the class

**Question No: 4 ( Marks: 1 ) - Please choose one**
How many elements are in the sample space of rolling one die:
**►6**
►12
►24
►36

**Question No: 5 ( Marks: 1 ) - Please choose one**
When two coins are tossed the probability of at most one head is:
►1/4
►2/4
**►3/4**
►4/4





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**Question No: 6 ( Marks: 1 ) - Please choose one**
If *A* and *B* are mutually exclusive events then *P*(*A* or *B*) equals:
► *P*(*A*) + *P*(*B*) - *P*(*A* and *B*)
► *P*(*A*) * *P*(*B*)
**► *P*(*A*) + *P*(*B*)  (Page 155)**
► *P*(*A*|*B*) + *P*(*B*|*A*)

**Question No: 7 ( Marks: 1 ) - Please choose one**
In scatter diagram the variable plotted along Y-axis is:
► Independent variable
**► Dependent variable**
► Any one
► Undefined

**Question No: 8 ( Marks: 1 ) - Please choose one**
Positive square root of variance of a distribution is:
► Rang
► Quartile deviation
**► Standard deviation  (Page 91)**
► only (a) &(c)

**Question No: 9 ( Marks: 1 ) - Please choose one**
When more values are lying at the start of the distribution, it is a:
► Symmetrical distribution
► Positively skewed
**► Negatively skewed**
► U shape figure

**Question No: 10 ( Marks: 1 ) - Please choose one**
What is ' $f_m$ ' in the formula of mode:

► First frequency
► Last frequency
**► Maximum frequency  (Page 54)**
► Minimum frequency

**Question No: 11 ( Marks: 1 ) - Please choose one**
If median = 7 and Mean = 5, what is the value of $Q_2$ :
►1
►3
►5
**►7  (because median = Q2)**

اطمینان قلب چاہتے ہو تو حسد سے دور رہو

**Question No: 12 ( Marks: 1 ) - Please choose one**
The probability of drawing a king of spade from a pack of 52 cards is:

► 1/4
► 1/13
► 1/26
**► 1/52**

**Question No: 13 ( Marks: 1 ) - Please choose one**
When referring to a curve whose longer tail is to the left, you would call it:

► U shape
**► Skewed to the left (Page 39)**
► Skewed to the right
► Symmetrical

**Question No: 14 ( Marks: 1 ) - Please choose one**
In statistics,we deal with:

► Individuals
► Isolated items
**► Aggregates of facts (Page 7) rep**
► Qualitative data

**Question No: 15 ( Marks: 1 ) - Please choose one**
When data is labeled to identify an attribute of element, the measurement scale is:

**► Ordinal (Page 9)**
► Interval
► Nominal
► Ratio

**Question No: 16 ( Marks: 1 ) - Please choose one**
The distribution is mesokurtic if the Moment Coefficient of kurtosis $b_2$ is:

► Equal to 0
**► Equal to 3**
► Less than 3
► Greater than 3

اس سے پہلے کہ تمہیں شہوت فتنے میں ڈالے نکاح کرلو

# MIDTERM EXAMINATION
## Spring 2009
## STA301- Statistics and Probability (Session - 2)

**Question No: 1 ( Marks: 1 ) - Please choose one**
A histogram consists of a set of adjacent rectangles whose bases are marked off by:
► **Class boundaries (Page 32)**
► Class limits
► Class marks
► Class frequency

**Question No: 2 ( Marks: 1 ) - Please choose one**
Stem and leaf was introduced by:
► Francis Galton
► Friedman
► **John Tukey (Page 47)**
► Pearson

**Question No: 3 ( Marks: 1 ) - Please choose one**
For positively skewed distribution
Mean…….Median…….Mode:
►=
►<
►> **Click here for detail**
► ≠

**Question No: 4 ( Marks: 1 ) - Please choose one**
For a positively skewed distribution:
► Mean<Mode<Median
► Mdian<Mode>Mean
► Mode>Mean>Median
► **Mean>Median>Mode** Click here for detail

**Question No: 5 ( Marks: 1 ) - Please choose one**
${}^{5}C_{5}$ Equals :
►1
►5
►10
►25

ہر چیز کی ایک پہچان ہوتی ہے اور عقلمند کی پہچان غور و فکر کرنا ہے اور غور و فکر کی پہچان خاموشی ہے


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**Question No: 6 ( Marks: 1 ) - Please choose one**
If a curve has a longer tail to the right, it is called :
► **Positively skewed (Page 39)**
► negatively skewed
► J-shaped
► Symmetric

**Question No: 7 ( Marks: 1 ) - Please choose one**
In measures of relative dispersion unit of measurement is:
► Changed
► **Vanish (Page 82)**
► Does not changed
► Dependent

**Question No: 8 ( Marks: 1 ) - Please choose one**
Data used by an agency which is originally collected by them is :
► **Primary data (Page 11)**
► Raw data
► Secondary data
► Grouped data

**Question No: 9 ( Marks: 1 ) - Please choose one**
When we toss a coin , we get only:
► **1 outcome**
► 2 outcomes
► 3 outcomes
► 4 outcomes

**Question No: 10 ( Marks: 1 ) - Please choose one**
Which of the following is the class frequency
► **The number of observations in each class**
► The difference between consecutive lower class limits
► Always contains at least 5 observations
► Usually a multiple of the lower limit of the first class

**Question No: 11 ( Marks: 1 ) - Please choose one**
Standard deviation is affected by the change of_____:
► Origin & scale
► Origin only
► **Scale only** Click here for detail
► Not origin & scale

اپنی مرضی اور اللہ کی مرضی میں فرق کا نام غم ہے

**Question No: 12 ( Marks: 1 ) - Please choose one**
For a moderately skewed distributions, the approximate percentage of case included between $\bar{X} - 2S$ *and* $\bar{X} + 2S$ is :
► 99.7%
► 68%
► **95%** **(Page 95)**
► 50%

**Question No: 13 ( Marks: 1 ) - Please choose one**
A die is rolled. What is the probability that the number rolled is greater than 2 and even:
► 1/2
► 1/3
► **2/3** **(greater than 2 = 3,4,5,6 = 4 numbers , 4/6=2/3)**
► 5/6

**Question No: 14 ( Marks: 1 ) - Please choose one**
The probability of drawing any one spade card is:
► 1/52
► 4/52
► **13/52**
► 52/52

**Question No: 15 ( Marks: 1 ) - Please choose one**
Which of the following is not the type of frequency curves?
► The symmetrical frequency curve
► The extremely skewed frequency curve
► The U-shaped frequency curve
► **Frequency polygon** **(Page 38)**

**Question No: 16 ( Marks: 1 ) - Please choose one**
If the third moment about mean is zero $(m_3 = 0)$ , then the distribution is:
► **Symmetrical**
► Negatively skewed
► Positively skewed
► Mesokurtic

وہ لوگ مبارک ہیں جو الفاظ سے نصیحت نہیں کرتے بلکہ عمل سے کرتے ہیں